REQUIREMENTS GATHERING DOCUMENT

Gaze Estimation for Student Attention Monitoring

1. Introduction

This document outlines the requirements for a Gaze Estimation system designed to monitor student attention in classroom or e-learning environments. The system uses computer vision and eye-tracking techniques to analyze where students are looking and determine their level of engagement.

2. Stakeholder Analysis

The following stakeholders have been identified along with their primary needs and interests in the system:

Stakeholder	Role	Primary Needs
Students	End users being monitored	Privacy assurance, minimal disruption to learning
Instructors / Teachers	Primary beneficiaries of insights	Real-time attention dashboards, engagement reports
School Administrators	Decision makers	Overall class performance, compliance with regulations
IT Department	System maintainers	Easy deployment, system stability, maintenance docs
Parents / Guardians	Indirect stakeholders	Transparency in data usage, student wellbeing

Stakeholder	Role	Primary Needs
Development Team	System builders	Clear and well-defined technical specifications

3. User Stories & Use Cases

3.1 User Stories

- As an instructor, I want to see which students are not paying attention so that I can re-engage them during the session.
- As an instructor, I want to receive a post-session report on student attention levels so that I can improve future lessons.
- As a student, I want to know the system is not recording me beyond eye movement data so that my privacy is respected.
- As an administrator, I want to view class-wide attention trends so that I can identify patterns across sessions.
- As an IT admin, I want the system to work with existing cameras so that no extra hardware is needed.

3.2 Use Cases

UC-01: Real-Time Attention Monitoring

- Actor: Instructor
- Description: The instructor views a live dashboard showing each student's attention status during an ongoing session.
- Precondition: Session has started and camera feed is active.
- Outcome: Instructor sees color-coded attention indicators per student.

UC-02: Session Report Generation

- Actor: Instructor / Administrator
- Description: After a session ends, the system generates an attention report with timestamps and attention scores.
- Precondition: Session data has been recorded.
- Outcome: A downloadable PDF/CSV report is available.

UC-03: Student Distraction Alert

- Actor: System (Automated)
- Description: The system detects a student has been looking away for more than a configurable threshold and triggers a soft alert.
- Precondition: Gaze tracking is active.

- Outcome: Alert is logged and optionally displayed to the instructor.

4. Functional Requirements

The following functional requirements define the core features and behaviors of the system:

ID	Requirement	Priority
FR-01	The system shall detect and track eye gaze direction in real time using a standard webcam.	High
FR-02	The system shall classify student attention as Focused, Distracted, or Away based on gaze direction.	High
FR-03	The system shall display a live dashboard showing attention status for each student.	High
FR-04	The system shall generate post-session attention reports exportable as PDF or CSV.	Medium
FR-05	The system shall allow instructors to configure distraction alert thresholds (e.g., 5–30 seconds).	Medium
FR-06	The system shall support multiple students simultaneously (up to 30 per session).	High
FR-07	The system shall allow instructors to start, pause, and end monitoring sessions.	High
FR-08	The system shall store session data securely with role-based access control.	High
FR-09	The system shall provide a student anonymization/privacy mode.	Medium
FR-10	The system shall support integration with common LMS platforms (e.g., Moodle, Google Classroom).	Low

5. Non-Functional Requirements

5.1 Performance

- The system shall process gaze detection at a minimum of 15 frames per second per student.
- The system shall provide attention status updates on the dashboard with no more than 2 seconds of delay.
- The system shall support at least 30 concurrent student streams without performance degradation.

5.2 Security

- All video and gaze data shall be encrypted in transit (TLS 1.2+) and at rest (AES-256).
- Access to session data and reports shall require role-based authentication.
- Raw video frames shall not be stored; only gaze coordinates and attention labels shall be persisted.
- The system shall comply with relevant data privacy regulations (e.g., GDPR, FERPA).

5.3 Usability

- The instructor dashboard shall be operable by a non-technical user with no more than 30 minutes of training.
- The interface shall be accessible on desktop browsers (Chrome, Firefox, Edge) without additional plugins.
- Key actions (start session, view alerts) shall be reachable within 2 clicks from the main screen.

5.4 Reliability

- The system shall maintain 99% uptime during scheduled class hours.
- In the event of a camera disconnect, the system shall gracefully handle the dropout and resume tracking when reconnected.
- Session data shall be auto-saved every 60 seconds to prevent loss due to unexpected failures.